



Automatic grading of human blastocysts from time-lapse imaging

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ABSTRACT

Background: Blastocyst morphology is a predictive marker for implantation success of in vitro fertilized human embryos. Morphology grading is therefore commonly used to select the embryo with the highest implantation potential. One of the challenges, however, is that morphology grading can be highly subjective when performed manually by embryologists. Grading systems generally discretize a continuous scale of low to high score, resulting in floating and unclear boundaries between grading categories. Manual annotations therefore suffer from large inter- and intra-observer variances.

Method: In this paper, we propose a method based on deep learning to automatically grade the morphological appearance of human blastocysts from time-lapse imaging. A convolutional neural network is trained to jointly predict inner cell mass (ICM) and trophectoderm (TE) grades from a single image frame, and a recurrent neural network is applied on top to incorporate temporal information of the expanding blastocysts from multiple frames.

Results: Results showed that the method achieved above human-level accuracies when evaluated on majority votes from an independent test set labeled by multiple embryologists. Furthermore, when evaluating implantation rates for embryos grouped by morphology grades, human embryologists and our method had a similar correlation between predicted embryo quality and pregnancy outcome.

Conclusions: The proposed method has shown improved performance of predicting ICM and TE grades on human blastocysts when utilizing temporal information available with time-lapse imaging. The algorithm is considered at least on par with human embryologists on quality estimation, as it performed better than the average human embryologist at ICM and TE prediction and provided a slightly better correlation between predicted embryo quality and implantability than human embryologists.

1. Introduction

Roughly 15% of the world population suffers from infertility problems, impeding or entirely preventing natural reproduction [1]. In vitro fertilization (IVF) is one of the most commonly and widely used fertility treatments. By fertilizing oocytes and culturing them outside the body for up to 6 days, optimal incubation conditions and intelligent selection of the best embryos for transfer can help overcome a number of infertility causes and thus significantly increase success rates.

A typical procedure starts by retrieving multiple oocytes from the ovaries of a woman followed by fertilization by IVF or intracytoplasmic sperm injection (ICSI). Zygotes are then cultured in an incubator under optimal conditions. With time-lapse imaging, the development and morphological appearance of the embryos can be monitored and evaluated, allowing only the most viable embryo(s) to be transferred to the uterus. Today, embryo evaluation is performed manually, typically by a single embryologist annotating morphological and kinetic

events (morphokinetics) such as blastocyst morphology and times of cell divisions up to blastocyst formation. The former comprises morphological grading of blastocyst expansion (BE), inner cell mass (ICM) and trophectoderm (TE) into a number of predefined categories of quality. Such morphokinetic events and morphological gradings have been shown to correlate well with implantation success and ultimately live birth [2,3].

However, manual annotation is tedious and prone to both inter- and intra-observer variance [4,5]. Especially morphological evaluation of embryos at the blastocyst stage has shown to be highly subjective [6,7]. Experiments with multiple embryologists annotating the same blastocysts thus show large disagreements, even with highly simplified grading systems. A number of possible reasons for the variations exist. Traditionally, blastocyst morphology grading is conducted as a static evaluation performed at a given point in time after blastocyst formation, typically around 115–120 h post insemination (hpi) [8,9].

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The exact evaluation time, however, is chosen by the embryologist and can easily affect the grade, as the blastocyst continues to evolve. For instance, a brief collapse of the blastocyst at observation time can significantly influence grading. Another factor is the possible use of grading scales as a local and relative comparison between a single patient's available embryos for transfer, instead of the intended use as a global comparison across patients. That is, some embryologists might use the system for ordering the available embryos of a single patient instead of grading them independently and objectively. This helps (local) differentiation, but violates the global definitions of quality described by the adopted grading system. Finally, another factor is that all blastocyst grading systems discretize continuous scales of morphological appearances into a number of predefined categories (e.g. low, medium, high quality). This makes the boundaries between categories both floating and unclear, resulting in a high amount of confusion between neighboring grades (e.g. A and B for the Gardner score [3]). Clinics and chains of clinics often establish consensus guidelines to try to overcome this issue. This requires periodic testing to identify potential grading deviations among and between embryologists. And naturally, the approach is expected to result in some inter-clinic variation.

By automating blastocyst grading using deterministic computer vision and machine learning algorithms, inter- and intra-observer variances can be completely eliminated. Furthermore, by showing a large number of biased examples (with observer variance) from embryologists, modern machine learning algorithms can potentially learn an unbiased representation of blastocyst morphology. The algorithm can thus learn to ignore local observer biases, while maintaining the relevant and necessary information from the embryologists' example annotations. Ideally, algorithms can be trained to surpass human performance in terms of both precision and speed.

In this paper, we propose a method based on deep learning to automatically predict ICM and TE grades on human blastocysts from time-lapse imaging. With time-lapse imaging, blastocyst expansion (BE) is implicitly represented by morphokinetic events and therefore often not annotated by embryologists. Therefore, in this study, BE is disregarded. Our method is based on a multi-task convolutional neural network operating on raw time-lapse images, generating visual image features for a recurrent neural network that predicts ICM grades (A, B, or C) and TE grades (A, B, or C) jointly. The method is fully automated and can be applied on all embryos. The main contributions of our work are:

1. A new approach to automate blastocyst morphology grading by incorporating temporal information available with time-lapse imaging.
2. Better accuracy of automated blastocyst morphology grading compared to the performance of human embryologists.
3. New customized loss function, Ordinal Cross-Entropy, that mixes classification and regression and outperforms common objective functions for both nominal and ordinal classification of blastocyst morphology.

1.1. Related work

For the past decade, a number of methods have been proposed to automate morphokinetic event detection and morphological grading of human embryos from time-lapse imaging during their early-stage development. Morphokinetic event detection involves identification of temporal events that have been found to correlate well with implantation success [2]. Among these are pronuclear appearance, cell divisions, and blastocyst formation. Especially the determination of cell division times by automatically counting blastomere cells in subsequent images has gained attention. Segmentation-based methods have investigated blastomere segmentation using either the shortest path approach on polar-converted image coordinates [10,11], or by comparing a number of generated cell hypotheses with an edge-filtered image to find the

most likely configuration [12–14]. Classification-based methods, on the other hand, directly estimate the number of cells by extracting a number of image features that are fed to a classifier trained for cell count prediction. Traditional classification methods use hand-crafted image features based on edges and texture [15–18], whereas more recent approaches, based on deep learning, learn both feature representations and classifier boundaries using end-to-end trainable convolutional neural networks (CNNs) [19–21]. Moreover, Rad et al. [22] combine segmentation and classification with deep learning by jointly estimating both cell counts and positions using a fully convolutional U-Net structure [23].

In addition to estimating cell counts from single images, some methods utilize temporal dependencies between neighboring frames. Ng et al. [20] investigate both early and late fusion of single-frame features before classification. Wang et al. [15] calculate an image similarity feature between subsequent frames, assuming large feature variations during cell divisions. Temporal consistency is then provided by globally maximizing single-frame cell predictions while ensuring monotonically increasing cell counts using either dynamic programming [15,20] or conditional random fields [11,19].

Whereas morphokinetic event detection involves the dynamic development of an embryo, morphological evaluation deals with the static appearance of an embryo at various development times. The final morphological evaluation happens at the blastocyst stage around day 5 after insemination. At this stage, the blastocyst expansion grade (BE) and morphological appearance of the inner cell mass (ICM) and the trophectoderm (TE) have been found to correlate well with implantation and are therefore often used for selecting which embryo to transfer [3]. ICM quality can be characterized by three grades: A (many cells, tightly packed), B (several cells, loosely grouped), and C (very few cells, disorganized). Similarly, TE quality can be characterized by three grades: A (many cells, forming a cohesive epithelium), B (few cells, forming a loose epithelium), and C (very few, large cells). Fig. 3 illustrates three examples of blastocysts with different ICM and TE grades.

Different methods have been proposed to segment both ICM and TE from single image frames. A common approach includes the use of different variants of the level set method to segment either ICM [24], TE [25], or both [26,27]. Kheradmand et al. [28], on the other hand, extract a number of hand-crafted features based on discrete cosine transform (DCT) coefficients from JPEG-encoded blocks and train a neural network to classify each block as either ICM, TE, cavity, zona pellucida, or background. Recently, deep learning-based methods have targeted ICM segmentation using fully convolutional neural networks such as FCN-32s [29] and U-Net [30]. These have both been reported to achieve superior performances to the level set methods mentioned above [30].

After segmenting ICM and TE, a number of features can be extracted from each region and fed to a classifier to predict blastocyst quality. Using this approach, Filho et al. [26] predict ICM and TE scores for human embryos according to the Gardner blastocyst grading system [3], whereas Rocha et al. [31] grade bovine embryos as either good, fair, or poor according to the International Embryo Technology Society (IETS) standard. The latter approach has since been applied on human embryos evaluated at 111.5 hpi, although on a very limited dataset [32,33]. However, predictions can also be performed without prior segmentation of ICM and TE. This has recently been demonstrated by Khosravi et al. [34] using a deep neural network to grade blastocysts as either good or poor from raw time-lapse images. The authors train a GoogLeNet Inception V1 model [35] on single focal planes taken at 110 hpi from the EmbryoScope® time-lapse system. They divide their dataset into good, fair, and poor quality embryos based on Veeck and Zaninovich grades annotated by embryologists. The authors, however, disregard the group of fair embryos (40% of the dataset) and thus train their network only on the best and worst quality blastocysts. The method achieves an accuracy of 97.5% on distinguishing embryos from

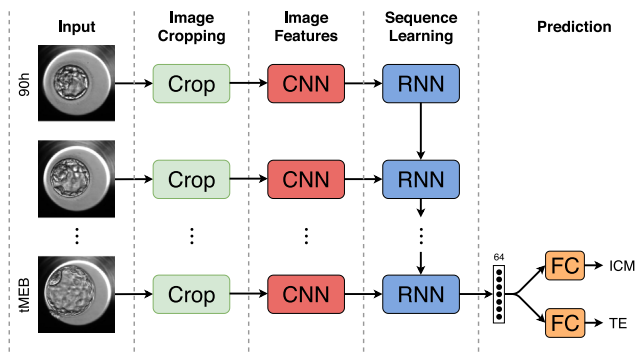


Fig. 1. System overview. Each image from the video sequence is fed through a cropper that extracts the embryo from the image and a CNN that extracts image features. An RNN utilizes temporal information by combining image features over time. Finally, two fully-connected (FC) layers use the 64 output features of the RNN to classify ICM and TE grades for the entire sequence.

the two groups on the test set and 90.4% accuracy on a smaller independent dataset with majority votes from five embryologists [34,36]. The results show that blastocyst quality can indeed be distinguished from raw time-lapse images. However, as all embryos in the fair category were disregarded during training, the algorithm has not seen fair embryos and is therefore not able to categorize these as being fair quality. Instead, the authors test their method on predicting successful pregnancy for all embryos regardless of quality, when combining good/poor predictions with age information in a decision tree.

In this body of work, we propose a fully automated procedure for predicting blastocyst grades of any embryo. A neural network jointly predicts grades for ICM (A, B, or C) and grades for TE (A, B, or C) based on raw time-lapse image sequences.

2. Methods

Fig. 1 presents a system overview of the automated system for predicting ICM and TE grades. A subset of image frames is first extracted from the available time-lapse sequence. The subset corresponds to the blastocyst stage of a developing embryo from 90 hpi to the time of maximum expansion. For each frame, 3 focal planes are then selected, generating an image with 3 channels. Based on this image, a cropper extracts a 224×224 pixel crop centered on the embryo (see [Appendix](#)), after which a CNN extracts image features. This is repeated for all frames in the extracted subsequence. The visual CNN features are then fed to a recurrent neural network (RNN) that connects subsequent frames from the sequence in order to leverage temporal information. Finally, two independent, fully-connected (FC) layers use the output features of the RNN to predict a single pair of ICM and TE grades for the entire sequence. The method thus utilizes a multi-task network structure, which often provides superior results for related tasks compared to individual classifiers trained separately for each task [37].

In the following subsections, the input generation, the CNN, and the RNN are each described in more detail. These are followed by the description of a customized loss function, which explicitly addresses the ordering of grades inherent in the A-B-C quality scale. A detailed description of the cropper is available in [Appendix](#).

2.1. Input generation

The input to the model illustrated in **Fig. 1** only covers a partial embryo development sequence. For each embryo, n , the input consists of frames from 90 hpi to the time of maximum expanded blastocyst (tMEB), sampled 1 h apart. tMEB is determined as the frame with the largest embryo area after 90 hpi using the embryo segmentation from

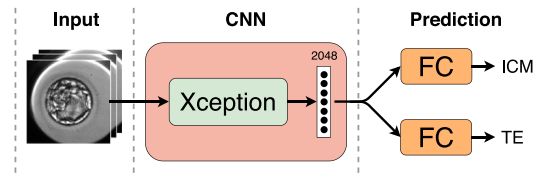


Fig. 2. Network architecture for CNN during training. A single cropped image with 3 focals (channels) is fed through a CNN based on the Xception architecture followed by two fully-connected (FC) layers classifying ICM and TE grades for the image.

Table 1

Network architecture for the CNN during training.

Layer	Output size	Parameters
Input	$224 \times 224 \times 3$	
Xception	1×2048	2M
ICM	1×3	6147
TE	1×3	6147

the cropper (see [Appendix](#)). Each frame consists of a number of focal planes, with the exact number depending on the specific time-lapse instrument used for data acquisition (see [Section 3](#)). Of the available 7–11 focal planes, 3 are selected, corresponding to the center focal ($0 \mu\text{m}$) and two peripheral focals ($-45 \mu\text{m}$ and $45 \mu\text{m}$). These are all similar across the different time-lapse instruments. In this way, a single model can be used across instruments, and the use of 3 focals allows us to fine-tune models pretrained on RGB images.

2.2. CNN

Each $224 \times 224 \times 3$ cropped input image is processed by a CNN as shown in **Fig. 2** to extract image features. For this, the Xception architecture [38] is used as a state-of-the-art backbone, by applying a global average pooling layer after the last convolutional layer. Effectively, this generates a 1×2048 feature vector for each image. During training, two separate fully-connected layers are used to classify ICM and TE grades jointly, allowing fine-tuning of the Xception network with pretrained ImageNet [39] weights. After training, the fully-connected layers are removed such that the CNN directly outputs image features. **Table 1** lists the layers as well as their output sizes and number of parameters.

For training the CNN, only a single image frame from each sequence was used. Here, different strategies were investigated for selecting which frame, such as the time of maximum expanded blastocyst (tMEB), or a fixed time after insemination such as 110 hpi or 115 hpi. However, the best results were obtained when using the annotation time chosen by the embryologist for each embryo. According to the guideline for blastocyst morphology grading with time-lapse, the annotation times are typically in the range 115–120 h post insemination (hpi) [9]. Henceforth, we refer to these embryo-specific annotation times as the annotation time of the embryologist (tA).

For inference, the CNN is applied across the entire image sequence. That is, even though the CNN was trained on only one frame from each sequence, during inference it is used to extract features for all frames in a sequence.

2.3. RNN

The RNN takes as input a sequence of CNN features of up to 30 frames from tMEB and backwards until 90 hpi, sampled 1 h apart. Trailing zero-vectors are used to pad blastocyst sequences shorter than 30 frames. Internally, the RNN consists of a single LSTM [40] block with 64 units and a bias vector. The hyperbolic tangent function is used for activation of the cell state, whereas a hard sigmoid is used for activation of the node output. During training, 50% input dropout

Table 2

Network architecture for the combined network. H and W denote the raw input height and width that both depend on the specific time-lapse instrument type (see Section 3).

Layer	Output size	Parameters
Input	$30 \times H \times W \times 3$	
Cropper	$30 \times 224 \times 224 \times 3$	469k
CNN	30×2048	2M
RNN	1×64	541k
ICM	1×3	195
TE	1×3	195

and 50% recurrent dropout are used. As for the CNN, the output of the RNN, a 1×64 feature vector, is fed to two separate fully-connected layers used to classify ICM and TE grades jointly. Fig. 1 illustrates the combined network, whereas Table 2 lists the layers as well as their output sizes and number of parameters.

2.4. Ordinal cross-entropy

As argued in the introduction, blastocyst morphology grading discretizes a continuous scale of morphological appearances into a number of predefined categories. In our case, the categories are A, B, and C for both ICM and TE. A traditional nominal classification network seeks to classify as many examples correctly, while disregarding the specific categories of incorrect predictions. That is, misclassification of an example with label A as either B or C results in the same error.

The task of grading blastocysts into a discrete number of categories, however, includes an implicit ordering of grades. Instead of nominal classification, this can be formulated as an ordinal regression problem. Ordinal regression is a mixture of regression and classification, as the objective is to predict a categorical label from an ordered set. A simple approach involves training a linear regression model by converting the class labels to positions on a linear, ordinal scale. Separate thresholds can then be fitted using the training data to define optimal intervals for each class [41]. Another approach, ordinal binary decomposition, decomposes the ordinal problem into multiple binary sub-problems [42]. In this paper, however, we propose an ordinal variation of the widely used categorical cross-entropy loss function.

For nominal multi-class classification, categorical cross-entropy is widely used to compare softmax outputs with one-hot encoded labels. The loss for a batch with N examples and C possible classes is defined as:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C y_{n,c} \log(\hat{y}_{n,c}) \quad (1)$$

where y denotes one-hot encoded ground truth labels and $\hat{y}$ denotes softmax predictions from the network. Assuming the correct category of a single example to be A, the one-hot encoded label is $y = [0, 0, 1]$. As categorical cross-entropy only focuses on the correct label, the two predictions $\hat{y}_1 = [0.1, 0.4, 0.5]$ and $\hat{y}_2 = [0.5, 0.0, 0.5]$ both result in the same loss of $L(y, \hat{y}_1) = L(y, \hat{y}_2) = 0.693$. The values of incorrect categories are thus not taken into account. However, for ranking problems such as blastocyst grading, $\hat{y}_1$ is generally considered a better prediction than $\hat{y}_2$, because the softmax probability distribution is more narrow around the correct category. That is, the uncertainty of $\hat{y}_1$ is closer to the correct category than the uncertainty of $\hat{y}_2$.

The proposed ordinal cross-entropy addresses this by focusing entirely on incorrect categories:

$$L(y, \hat{y}) = -\frac{1}{N} \sum_{n=1}^N \sum_{c=1}^C (1 - y_{n,c}) \log(1 - \hat{y}_{n,c}) w(c, y_n) \quad (2)$$

where $w(c, y_n)$ is a weighting function with y_n denoting the one-hot encoded ground truth vector for the example n . Assuming equal weights for all classes, $w(c, y_n) = 1$, the same predictions as above result in two

different losses, namely $L(y, \hat{y}_1) = 0.616$ and $L(y, \hat{y}_2) = 0.693$. This is due to the fact that the distribution of incorrect predictions is taken into account. If we further define the weight function as the euclidean distance between the evaluated class, c , and the ground truth class, $\text{argmax } y_n$:

$$w(c, y_n) = |c - \text{argmax } y_n|, \quad (3)$$

the two losses are $L(y, \hat{y}_1) = 0.722$ and $L(y, \hat{y}_2) = 1.386$. Ultimately, the ordinal cross-entropy loss function considers the order of predictions and thus ‘awards’ the first prediction to be closer to the desired label than the second. By ignoring the predicted value of the correct category, the loss function effectively penalizes incorrect predictions instead of awarding the correct ones.

2.5. Training procedure

The CNN and RNN were trained separately using the training set consisting of 6957 embryos, each with a single ICM and TE annotation. To compensate for class imbalance, an additional weight term was included in the loss function to make infrequent class labels have more impact than frequent class labels. The distribution of A/B/C scores across the training set were 47%/40%/13% and 43%/40%/17% for ICM and TE, respectively. The loss for each example was therefore weighted by inverse label frequencies (e.g. $100\%/13\% = 5.9$ for ICM grade C), effectively weighting infrequent class labels higher than frequent class labels. ICM and TE loss contributions were weighted equally. Both networks were trained separately, each for 30 epochs with the Adam optimizer on a GeForce Titan X GPU with 12 GB RAM running TensorFlow version 1.8.0. After each epoch, an average loss over the validation set was computed, and the model with the smallest validation loss across the 30 epochs was used for evaluating the test set. The CNN was trained with a batch size of 24, whereas a batch size of 32 was used for the RNN. For both models, the learning rate was initially set to 0.001 and continually reduced by a factor of 2 when the validation loss did not drop for 5 consecutive epochs. Experiments including combined training of the CNN and RNN were carried out, but without experiencing any performance increases over the separate training strategy described above.

For the CNN, two data augmentation strategies were used during training. The selection of 3 focal planes described in Section 2.1 was randomly shifted ± 1 focals ($\pm 15 \mu\text{m}$) to increase invariance towards imperfect focus. Secondly, the time of the selected image frame was randomly shifted by up to ± 30 minutes to increase invariance towards embryo development speed and imprecise annotation times. The same data augmentation was applied across all CNN configurations regardless of the frame selection strategy (e.g. 110 hpi, tMEB, or tA). No additional data augmentation was applied during training of the RNN.

3. Data

To train and evaluate the proposed CNN and RNN, a large dataset (dataset 1) was collected from four different fertility clinics. The dataset consists of 4032 individual treatments undergoing IVF (1169), intracytoplasmic sperm injection (ICSI) (2534), or a mixture of both treatments (329) with an average patient age of 35.0 years. From these treatments, 8664 embryos that had reached the blastocyst stage were analyzed retrospectively. The dataset has been randomly split into three subsets for training (80%), validation (10%), and testing (10%).

Of the 8664 embryos, 4483 were cultured and recorded in an EmbryoScope®+ time-lapse system. The system includes a custom-made microscope with $16\times$ magnification and a 0.50 LWD Hoffman modulation contrast objective. For each embryo, the system provides 11 focal planes with $15 \mu\text{m}$ separation every 10 min. Each focal plane is a monochrome 8-bit image of size 800×800 with a resolution of 2.9 pixels per μm . The remaining 4181 embryos were cultured and recorded in an EmbryoScope® time-lapse system. The system includes a

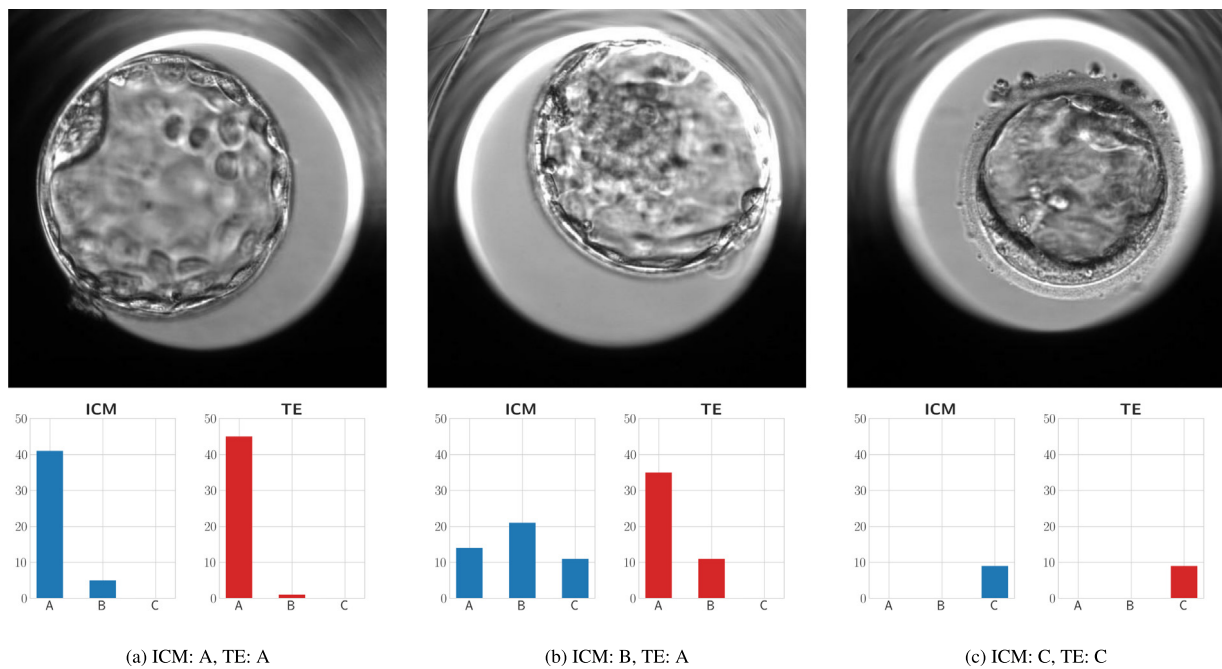


Fig. 3. Example blastocysts from dataset 2 with different ICM and TE ratings. The upper row shows the center focal planes at 115 hpi, while the lower row shows histograms of the embryologists' annotations.

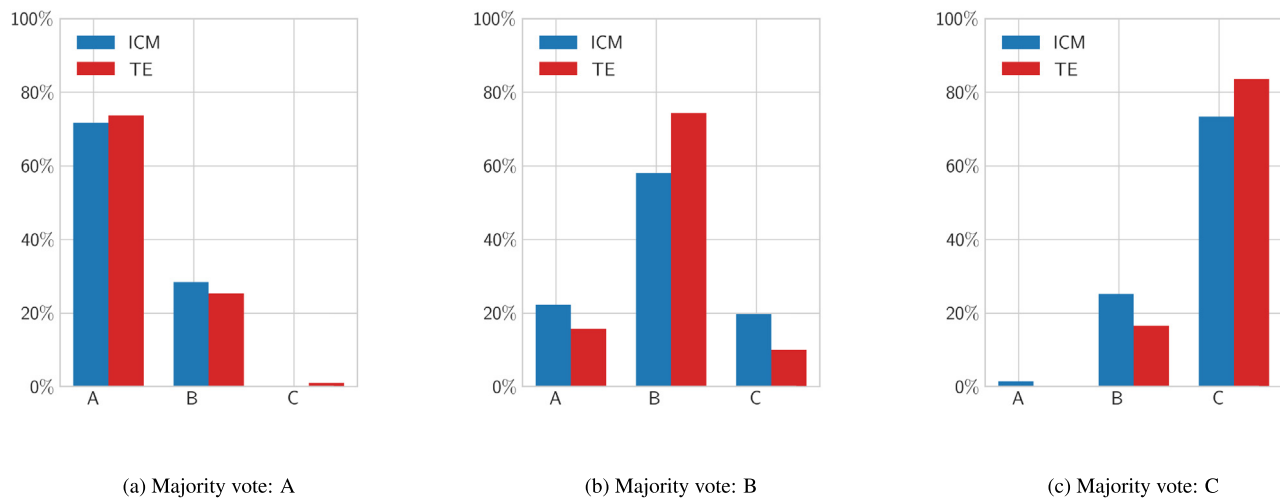


Fig. 4. Inter-rater agreement on ICM and TE annotations from dataset 2. (a) shows the distribution of ICM/TE annotations across multiple embryologists for examples where the majority of the embryologists labeled ICM/TE as A. (b) and (c) illustrate the same distributions for examples with majority votes B and C, respectively.

Leica custom-made microscope with 20 $\times$ magnification and a 0.40 LWD Hoffman modulation contrast objective. For each embryo, the system provides 7 focal planes with 15 μ m separation every 10–15 min. Each focal plane is a monochrome 8-bit image of size 500 $\times$ 500 with a resolution of 1.7 pixels per μ m.

All embryos in dataset 1 were annotated once with ICM grades (A, B, or C) and TE grades (A, B, or C) by experienced embryologists from the clinics. The annotations were conducted according to the guidelines from Vitrolife [9], which generalize the Gardner blastocyst grading system [3] to time-lapse imaging. The distribution of A/B/C scores for all data subsets (training, validation, and test) were 47%/40%/13% and 43%/40%/17% for ICM and TE, respectively. The distribution for the training set was used to compensate for class imbalance as explained in Section 2.5.

As explained in the introduction, manual annotations suffer from large inter- and intra-observer variances. Dataset 1 was thus expected to contain annotator biases that would influence both training and evaluation of an automated algorithm. Therefore, for evaluation purposes only, a smaller, independent test set (dataset 2) of 55 embryos cultured and recorded in an EmbryoScope[®] time-lapse system was annotated by multiple embryologists. Each embryo was annotated by between 5 and 46 embryologists. Fig. 3 shows three examples of blastocysts (center focal plane) along with histograms showing embryologist annotations. In Fig. 3(a), a large majority of embryologists labeled the blastocyst as A for both ICM and TE, whereas only a few embryologists labeled either ICM or TE as B. This example shows a large agreement between embryologists. Fig. 3(b), however, illustrates a smaller agreement, as only a small majority of embryologists labeled ICM as B over A and C. Fig. 3(c), ultimately, shows how examples that were difficult to grade

Table 3

Results on test set (dataset 1) for various models.

Model	Frames	Accuracy		MSE	
		ICM	TE	ICM	TE
CNN ($t = 105$ hpi)	1	0.576	0.603	0.491	0.454
CNN ($t = 110$ hpi)	1	0.572	0.622	0.555	0.459
CNN ($t = 115$ hpi)	1	0.586	0.611	0.533	0.474
CNN ($t = 120$ hpi)	1	0.580	0.645	0.525	0.411
CNN ($t = \text{tMEB}$)	1	0.577	0.656	0.476	0.387
CNN ($t = \text{tA}$)	1	0.639	0.652	0.417	0.383
RNN	30	0.652	0.696	0.415	0.361

were only labeled by a small amount of embryologists, whereas the majority (37 of 46 embryologists) chose not to annotate or assign a grade.

Fig. 4 illustrates label agreements across the entire multi-annotator test set (dataset 2) grouped by majority votes for ICM and TE, respectively. Fig. 4(a) thus shows annotation histograms for ICM and TE, where the majority of embryologists labeled the examples as A. For these examples, some embryologists labeled B instead of A, whereas there were two occurrences of an embryologist labeling C instead of A. The histograms generally show that confusion between immediately adjacent labels (e.g. C and B) was common, whereas confusion between C and A (and vice versa) was rare. Moreover, they generally show slightly larger agreements between embryologists on TE compared to ICM.

4. Results

To evaluate the proposed method, three different experiments were carried out on dataset 1 presented in Section 3. The experiments are presented in the following three subsections. In the first section, different variants of both the CNN and RNN are presented and evaluated on the test set (dataset 1) consisting of 851 embryos. Of these, the best model is compared to human annotation performance on the independent multi-annotator test set (dataset 2) with 55 embryos. In the second section, the proposed ordinal cross-entropy loss function is evaluated against nominal classification and two variants of ordinal regression. Finally, in the third section, network predictions are compared to human annotations in terms of implantation rates for embryos with known pregnancy outcome.

4.1. Results overview

As described in Section 1.1, existing methods for blastocyst quality prediction are all based on static images, acquired at a fixed time after insemination [31,33,34]. To compare with our approach, we trained six separate CNNs, using single image frames extracted either at fixed times (105, 110, 115, and 120 hpi), at tMEB, or at tA. The CNN trained at tA was further used as feature extractor for training an RNN as described in Section 2.3. Table 3 presents accuracies and mean squared errors (MSE) for each model, evaluated on ICM and TE grading independently. For calculating MSE, the categories A, B, and C were translated to positions on a linear, ordinal scale ($C = 0$, $B = 1$, $A = 2$).

From Table 3, it is clear that the RNN, using spatial and temporal information from 30 frames, achieved the best results in terms of both accuracy and MSE. Moreover, fixing the evaluation (and training) time to 105, 110, 115, or 120 hpi provided inferior results compared to adapting the evaluation time to the embryo development stage (tMEB, tA). Table 5 presents confusion matrices for the RNN evaluated on ICM and TE, individually. Clearly, the network has learned the ordinal scale of the data, with considerably less confusion between A and C than between A and B, and B and C. A/C disagreements with embryologists thus amounted to only 2.2% and 1.9% for ICM and TE, respectively.

As mentioned in Section 2, the proposed method utilizes a multi-task network structure. Table 4 compares this approach with individual

Table 4

Results on test set (dataset 1) comparing multi-task model vs. individual models for predicting ICM and TE. To ease comparison, CNN and RNN results for the multi-task model were copied from Table 3.

Model	Accuracy		MSE	
	ICM	TE	ICM	TE
CNN (ICM)	0.622	–	0.431	–
CNN (TE)	–	0.624	–	0.422
CNN (multi-task)	0.639	0.652	0.417	0.383
RNN (ICM)	0.645	–	0.411	–
RNN (TE)	–	0.680	–	0.358
RNN (multi-task)	0.652	0.696	0.415	0.361

Table 5Confusion matrices for ICM and TE showing relation between network (RNN) predictions and single human annotations on the test set (dataset 1, $n = 851$).

(a) ICM

		Network		
		C	B	A
Annotator	C	69	24	5
	B	68	193	81
	A	14	104	293

(b) TE

		Network		
		C	B	A
Annotator	C	112	28	4
	B	51	216	67
	A	12	97	264

Table 6Confusion matrices for ICM and TE showing relation between network predictions and majority votes from multiple human annotators on the multi-annotator test set (dataset 2, $n = 55$).

(a) ICM

		Network		
		C	B	A
Majority vote	C	2	7	1
	B	1	22	6
	A	0	4	12

(b) TE

		Network		
		C	B	A
Majority vote	C	4	4	0
	B	0	21	4
	A	0	7	15

CNN and RNN models trained for ICM and TE prediction, separately. The multi-task model provided the best accuracies on both CNN and RNN levels. Although MSE was slightly better for the individual models, the multi-task model was generally considered superior. Moreover, a combined model practically halves the computation time during both training and runtime, making it favorable for potential use in production.

4.1.1. Human vs. network

For evaluating the performance of the RNN vs. human embryologists, the independent multi-annotator dataset 2 with 55 embryos was used. Table 6 presents the confusion matrices for the RNN evaluated on ICM and TE, individually, against majority votes from all embryologists. Accuracies of 65.5% and 72.7% were achieved for ICM and TE, respectively.

For a fair and unbiased comparison, leave-one-annotator-out cross-validation was applied on each example similar to the approach by Steidl et al. [43]. Let N denote the number of examples, A_n denote the number of annotators with valid annotations for the example n , y_n^a denote the a 'th annotator's annotation for example n , and $y_n^{\bar{a}} = [y_n^1, \dots, y_n^{a-1}, y_n^{a+1}, \dots, y_n^{A_n}]$ denote the vector of annotations from the remaining annotators of example n . Then, the average per-example accuracy for human embryologists can be calculated as:

$$\text{accuracy}_{\text{human}}(y) = \frac{1}{n} \sum_{n=1}^N \frac{1}{A_n} \sum_{a=1}^{A_n} \mathbf{1}[\text{vote}(y_n^{\bar{a}}) = y_n^a] \quad (4)$$

Table 7

CNN results on test set (dataset 1) for different loss functions. Results for ordinal cross-entropy are copied from Table 3 for ease of comparison.

Loss function	Accuracy		MSE	
	ICM	TE	ICM	TE
Linear regression (thresholded)	0.526	0.519	0.664	0.727
Categorical cross-entropy	0.610	0.644	0.503	0.444
Ordinal binary decomposition	0.589	0.617	0.475	0.464
Ordinal cross-entropy (proposed)	0.639	0.652	0.417	0.383

where $\mathbf{1}[x]$ is the indicator function and $\text{vote}(x)$ is a voting function, returning the most frequently occurring annotation from the list. Similarly, letting $\hat{y}_n$ denote the network prediction for example n , the average per-example accuracy for the network can be calculated as:

$$\text{accuracy}_{\text{network}}(y, \hat{y}) = \frac{1}{n} \sum_{n=1}^N \frac{1}{A_n} \sum_{a=1}^{A_n} \mathbf{1}[\text{vote}(y_n^a) = \hat{y}_n] \quad (5)$$

Applying Eqs. (4) and (5) to the test set (dataset 2) of 55 embryos, human embryologists achieved ICM and TE accuracies of 65.1% and 73.8%, whereas the network reached 71.9% and 76.4%. That is, in average, the network performed better than individual human embryologists in predicting majority votes on both ICM and TE.

4.2. Loss functions

To evaluate the proposed ordinal cross-entropy loss function, we compared its performance against three other loss functions described in Section 2.4. Table 7 lists results for the CNN ($t = \text{tA}$) trained with either of these. The proposed ordinal cross-entropy provided the best results, both in terms of accuracy and MSE. As described in Section 2.4, the main purpose of the proposed loss function was to minimize confusion between A and C using the ordinal weight function in Eq. (3). This corresponds to minimizing the mean absolute error, which is tightly coupled to MSE. However, when comparing with categorical cross-entropy, not only MSE was improved. The accuracy for ordinal cross-entropy was 2.9% and 0.8% higher than categorical cross-entropy for ICM and TE, respectively. This supports the general claim that ordinal methods often improve accuracy while minimizing distances [41].

4.3. Implantation rate

As stated in the introduction, blastocyst morphology grading is commonly used for embryo selection, as the grades of both ICM and TE have been shown to correlate well with actual implantation rates. As part of the test set (dataset 1) had annotations of implantation (fetal heartbeat), a comparison can be made of how well morphological scores given by embryologists or the network correlate with implantation. Of the 851 embryos in the test set, 287 were transferred and annotated with implantation. Of these, 87 had fetal heartbeat and 200 did not. For the evaluation, we divided blastocysts into three groups based on ICM and TE grades. Top quality blastocysts (TQB) consisted of embryos with ICM/TE grade A/A. Good quality blastocysts (GQB) consisted of embryos with grades A/B, B/A, and B/B. And finally, poor quality blastocysts (PQB) consisted of embryos with remaining grades C/A, A/C, C/B, B/C, and C/C.

Receiver operating characteristic (ROC) curves were generated by gradually decreasing the quality threshold from TQB to PQB. For each threshold, the model's ability to distinguish positive and negative fetal heartbeats were calculated in terms of a true positive rate and a false positive rate. Fig. 5 illustrates the resulting ROC curves for both human embryologists and the network. In order to aggregate correlation results across the three quality groups, an area under the curve (AUC) was calculated for each ROC curve. Human embryologists achieved an AUC of 0.64, whereas the network achieved an AUC of 0.66. However, no significant difference was found between the two AUCs ($p = 0.536$) [44].

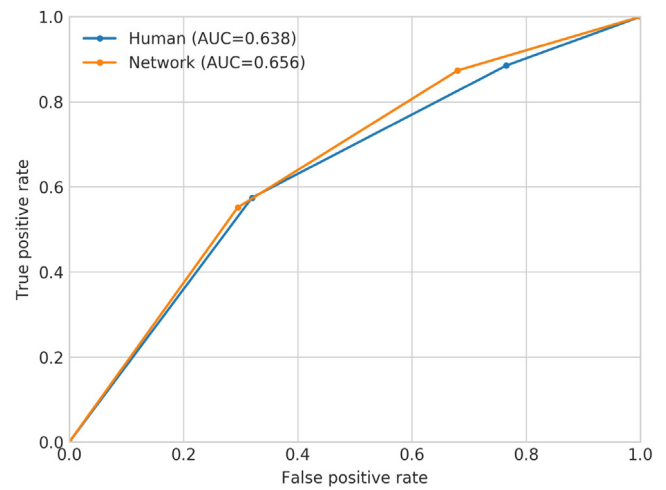


Fig. 5. ROC curves for human embryologists and the network.

Table 8

Average computation times for the CNN and RNN in order to predict ICM and TE grades for a single blastocyst. The timing was performed on a workstation with Intel Xeon E5-2620 v4 CPU and Nvidia GeForce Titan X GPU.

Platform	CNN (sec)	RNN (sec)	Total (sec)
CPU	5.938	0.026	5.964
GPU	0.548	0.048	0.596

4.4. Timing

The proposed method was trained on a dedicated GPU-server. For runtime inference, however, the model can be executed on either CPU or GPU. Table 8 lists average computation times for the CNN (for 30 frames) and RNN. The total execution time denotes the processing time for a single blastocyst. Using GPU acceleration, ICM and TE predictions of a blastocyst can thus be carried out in approximately 0.6 s.

5. Discussion

The above results suggest that the network is at least on par with human performance on blastocyst grading. As such, it performed better than individual human embryologists in predicting majority votes on both ICM and TE on dataset 2 with 55 embryos annotated by multiple embryologists. When evaluating the relationship between morphology grades and implantation, the network provided a slightly (but not significantly) better correlation than human embryologists between predicted embryo quality and implantability. The automated grouping of embryos into either TQB, GQB, or PQB is therefore considered at least as good as the manual equivalent of human embryologists for selecting the most viable embryos.

The proposed algorithm is fully deterministic and will therefore always provide consistent predictions with no inter- or intra-observer variances. This makes it ideal for applying subsequent global implantation models such as the KIDScore D5 model [45] that combines and translates morphokinetic and morphological annotations on day 5 after insemination into embryo viability scores. The algorithm is furthermore fully automated and requires no manual pre-filtering of embryos.

Existing state-of-the-art algorithms for blastocyst quality assessment all base their predictions on static images, acquired at a fixed time after insemination [31,33,34]. In this paper, however, we have shown that the use of temporal information from time-lapse imaging can indeed improve performance. As such, combining information from 30 image frames between 90 hpi and tMEB, our RNN achieved accuracy improvements of 7.2% and 5.1% on ICM and TE, respectively, compared to the performance of a single-frame CNN applied to static images

taken at 120 hpi. Although no direct comparison can be made to related methods due to different datasets and different objectives, our results on A/C disagreements as mentioned in Section 4.1 may relate to the objective of distinguishing good and poor embryos in Khosravi et al. [34]. Disregarding the group of fair embryos [34] may thus be similar to disregarding ICM and TE annotations with grade B. In this case, the reported accuracy of 97.5% by Khosravi et al. [34] may be comparable to our accuracies of distinguishing A and C of 97.8% (2.2% disagreement) for ICM and 98.1% (1.9% disagreement) for TE.

For all trained models in Table 3, the accuracy of TE was well beyond the accuracy of ICM. That is, the algorithms were generally better at predicting TE than ICM. This follows the trend from Fig. 4 that showed larger agreements between embryologists on TE grades compared to ICM grades. The average agreements from Fig. 4 may in fact indicate an upper limit on the obtainable accuracy for ICM and TE, individually, when evaluated on single human annotations. In future work, more emphasis should therefore be put on evaluating large multi-annotator datasets, as majority votes allow inter-rater variation to be reduced significantly. Ultimately, a network trained on a large amount of majority votes could surpass human performance by a large margin. However, acquiring such a dataset is both costly and time-consuming, as clinics in practice only annotate each embryo once.

Section 4.3 evaluated the correlation between predicted embryo grades (ICM and TE) and pregnancy outcome and showed no significant difference between human and network predictions. Training on majority votes from multiple embryologists might reveal a greater difference in favor of the network. However, to completely eliminate subjectivity and inter- and intra-rater variation during embryo grading, models could be trained directly at predicting pregnancy outcome. Tran et al. [46] has shown promising results in this context and reported an AUC of 0.93 for distinguishing which embryos that resulted in a fetal heartbeat and which embryos that were either discarded manually by embryologists or did not result in a fetal heartbeat. As only a small subset of the present dataset includes known outcome, however, a comparison with such methods is beyond the scope of this paper.

In future work, the proposed method should be validated using leave-one-clinic-out cross-validation. This would help provide an unbiased estimate of the capability of the model to generalize to new clinics. The current dataset, however, consists of only four clinics, each with a different distribution of embryos from the two time-lapse instruments, and each with different distributions of A/B/C scores. Therefore, leaving a clinic entirely out during training leaves the training set unbalanced and thus introduces clinic-specific biases into the trained model. Adding more clinics to the dataset would reduce such biases and allow for reliable leave-one-out cross validation.

6. Conclusion

This paper has presented a method based on deep learning to fully automate blastocyst morphology grading of inner cell mass (ICM) and trophectoderm (TE) from time-lapse imaging of human embryos. The method utilizes a deep neural network applied on sequences of images, utilizing both spatial and temporal information for predicting blastocyst quality.

Results have shown that the addition of temporal data using time-lapse imaging provided considerably better results than a static, single-frame evaluation performed at a fixed time after insemination. Evaluating classification performance of human embryologists vs. the proposed method on majority votes from an independent multi-annotator test set, the method outperformed the average human embryologist on both predicting both ICM and TE grades. Ultimately, by grouping blastocyst morphology predictions into three quality groups, ROC AUC's were calculated for embryos with known pregnancy outcome. Here, the proposed method provided a similar correlation between the predicted embryo quality and the pregnancy outcome to human embryologists.

Table A.9

U-Net network architecture for cropper. Some operations are abbreviated: convolution (conv), rectified linear unit (ReLU), batch normalization (BN), max pooling (pool), and nearest neighbor upsampling (upsample).

Layer	Output size	Operations
Input	$64 \times 64 \times 3$	
Contraction block 1 (CB1)	$64 \times 64 \times 32$	$[3 \times 3 \text{ conv} + \text{ReLU} + \text{BN}] \times 2$
Max pool	$32 \times 32 \times 32$	$2 \times 2 \text{ pool}$
Contraction block 2 (CB2)	$32 \times 32 \times 64$	$[3 \times 3 \text{ conv} + \text{ReLU} + \text{BN}] \times 2$
Max pool	$16 \times 16 \times 64$	$2 \times 2 \text{ pool}$
Bottleneck	$16 \times 16 \times 128$	$[3 \times 3 \text{ conv} + \text{ReLU} + \text{BN}] \times 2$
Upsample	$32 \times 32 \times 64$	$2 \times 2 \text{ upsample} + 2 \times 2 \text{ conv}$
Concatenate	$32 \times 32 \times 128$	Concatenation with CB2
Expansion block 1 (EB1)	$32 \times 32 \times 64$	$[3 \times 3 \text{ conv} + \text{ReLU} + \text{BN}] \times 2$
Upsample	$64 \times 64 \times 32$	$2 \times 2 \text{ upsample} + 2 \times 2 \text{ conv}$
Concatenate	$64 \times 64 \times 64$	Concatenation with CB1
Expansion block 2 (EB2)	$64 \times 64 \times 32$	$[3 \times 3 \text{ conv} + \text{ReLU} + \text{BN}] \times 2$
Classification	$64 \times 64 \times 1$	$1 \times 1 \text{ conv}$

In summary, the results suggest that the proposed method is at least on par with and possibly superior to the performance of a human embryologist. Requiring no manual pre-filtering of embryos, the method is fully automated and thus time-efficient in a busy IVF laboratory. And due to the fully deterministic algorithm, it ensures consistent predictions with no inter- or intra-observer variances.

Declaration of competing interest

Mikkel F. Kragh, Jens Rimestad, and Jørgen Berntsen are all employed by Vitrolife A/S. Jørgen Berntsen and Jens Rimestad further have stock ownership in Vitrolife A/S.

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Appendix. Cropper

In order to reduce computational complexity while optimizing the image resolution for the CNN, we crop out 64% of the image (80% along each axis) centered on the embryo. A crop of 64% removes a considerable part of the image that does not contain embryo pixels, while ensuring that even the largest blastocysts are not truncated.

For centering the crop on the embryo, a small U-Net [23] segmentation network is trained for pixel-wise binary classification of embryo vs. non-embryo. Table A.9 lists all layers in the network, the output size of each layer, and the operations involved. The network is provided a downscaled embryo image of size $64 \times 64 \times 3$ (3 focal planes) and outputs a binary segmentation mask of size $64 \times 64 \times 1$. From the segmentation mask, the center of mass of all predicted embryo-pixels is then used to center a 64% crop in the original size image on the embryo. This results in images of size 640×640 for EmbryoScope[®]+ images and 400×400 for EmbryoScope[®] images. Finally, the crop is resized to 224×224 pixels for all images, such that the input is consistent and comparable across the different time-lapse instruments. Furthermore, a resolution of exactly 224×224 pixels allows us to use pretrained ImageNet [39] weights that are readily available for most state-of-the-art network architectures.

To train and test the cropper, a dataset of 2126 time-lapse images was collected and annotated manually as either embryo or non-embryo for each pixel. 1119 images were recorded in an EmbryoScope[®]+ time-lapse system, whereas 1007 were recorded in an EmbryoScope[®] time-lapse system. 1509 contained embryos at different development stages, whereas 617 represented wells without an embryo. The dataset was randomly split into two subsets for training (80%) and testing (20%).

Random horizontal flipping, $\pm 15^\circ$ rotation, $\pm 10\%$ horizontal and vertical shift, and zoom in the range [0.8, 1.1] was applied to each image for data augmentation. The network was trained over 200 epochs with a batch size of 16 using the Adam optimizer [47] with an initial learning rate of 0.5. The Dice coefficient was used as a loss function.

Evaluated on the test set of 425 embryos, the cropper achieved a Dice coefficient of 0.960. A score close to 1 indicates a large overlap between predicted and ground truth embryo masks.

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